

Study Report: Python Programming for Beginners - Basic Language from Absolute Beginners to Intermediate

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Family: Python Beginners Core

Level: Beginner

Chapters: 14

Source: Python Programming for Beginners - Basic Language from Absolute Beginners to Intermediate.epub

Summary

This report covers a book that takes readers from absolute beginner to intermediate. It presents Python and its uses, first steps and basics, using Python as a programmable calculator, dictionaries and data structuring, strings, reading and writing files, organizing files, debugging tools, tuples, web scraping, data and statistics management, and working with Excel charts, spreadsheets, and images. It balances core programming with practical, real-world tasks.

Chapters

Chapter 1: Presentation of Python

Chapter focus: Presentation of Python

Python is a general-purpose programming language known for readable syntax and wide use in web development, automation, data analysis, and machine learning. Unlike many languages that use braces and semicolons, Python uses indentation to show structure, which helps beginners see how code is organized. You write instructions in a .py file or run them line by line in the interactive shell. Each instruction tells the computer to do something: store a value, print text, make a decision, or repeat a task.

Code Reference:

```
print("Hello, Python!")  
print(2 + 3)
```

What it shows: The first line prints text. The second shows Python can also calculate numbers.

Topic guide

What it actually is

Python is a high-level, interpreted language: you write human-readable code and the Python interpreter runs it without a separate compile step. It comes with a large standard library and thousands of third-party

packages.

When to use it

Use Python when you want to build something quickly, automate repetitive tasks, analyze data, create a web backend, or learn programming for the first time.

Where to use it

Used at Google, Netflix, NASA, and in schools worldwide. Common in data science teams, DevOps automation, scripting, and beginner coding courses.

Real use example

A small business owner writes a 10-line Python script that renames 200 invoice PDFs by date every month, saving an hour of manual work.

Chapter 2: The Language of Python

Chapter focus: The Language of Python

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Chapter 3: First Steps with Python

Chapter focus: First Steps with Python

Before writing Python programs, you install the Python interpreter from python.org (choose Python 3). During setup on Windows, check 'Add Python to PATH' so you can run python from the terminal. After installation, open a terminal and type `python --version` to confirm. You can write code in any text editor, but beginners often use IDLE (included with Python) or VS Code. The interactive shell (`>>>` prompt) is useful for testing one line at a time; saved `.py` files are for complete programs.

Code Reference:

```
import sys
print(sys.version)
print("Setup OK")
```

What it shows: import sys loads system info; sys.version shows your Python version.

Topic guide

What it actually is

Installation means putting the Python interpreter and tools on your computer so it can read and execute `.py` files. PATH is a list of folders Windows/Mac search when you type a command.

When to use it

Install Python once on any machine where you plan to write or run Python code. Reinstall or upgrade when you need a newer version for a library or course.

Where to use it

On your laptop for learning, on a server for web apps, on a Raspberry Pi for hardware projects, or in cloud notebooks like Google Colab (no local install needed there).

Real use example

A student installs Python 3.12, opens VS Code, creates `hello.py`, runs it from the terminal, and sees 'Setup OK' - confirming the environment works before starting homework.

Chapter 4: The Basics of Python

Chapter focus: The Basics of Python

A variable is a name that points to a value stored in memory. You create one with the assignment operator (`=`): the name goes on the left, the value on the right. Python figures out the type automatically - you do not declare `int` or `str` first. Common types are integers (whole numbers), floats (decimals), strings (text in quotes), and booleans (`True/False`). You can change a variable later by assigning a new value. Clear names like `user_age` or `total_price` make code easier to read than single letters.

Code Reference:

```
count = 5          # integer
```

```
label = "box"      # string
price = 2.5        # float
is_open = True     # boolean
print(type(count)) # <class 'int'>
```

What it shows: Each variable stores one value. type() tells you what kind of data it holds.

Topic guide

What it actually is

Variables are labeled containers for data. The label (name) lets you reuse and update values throughout a program without repeating the actual number or text.

When to use it

Use variables whenever a value might change, will be reused later, or needs a meaningful name - scores, user names, prices, flags, counters.

Where to use it

Every Python program: games (player score), web apps (username), data scripts (file path), calculators (operands), and loops (counters).

Real use example

An online shop stores item_price = 499, quantity = 2, and computes total = item_price * quantity so the checkout page shows Rs 998 without hard-coding numbers.

Chapter 5: Python as a Programmable Calculator

Chapter focus: Python as a Programmable Calculator

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An online shop stores `item_price = 499`, `quantity = 2`, and computes `total = item_price * quantity` so the checkout page shows Rs 998 without hard-coding numbers.

Chapter 6: Dictionaries and Data Structuring

Chapter focus: Dictionaries and Data Structuring

Tuples use parentheses and cannot change after creation - good for fixed data like coordinates. Sets use curly braces and keep only unique items, with no guaranteed order. Dictionaries map keys to values (`{"name": "Ana", "age": 14}`) for fast lookup by label. Choose a tuple when data must stay fixed, a set when you need uniqueness, and a dict when each value has a meaningful name.

Code Reference:

```
point = (10, 20)
unique = {1, 2, 2, 3}
user = {"name": "Ana", "role": "admin"}
print(point[0], unique, user["name"])
```

What it shows: Tuple index works; set removes duplicate 2; dict lookup uses the key 'name'.

Topic guide

What it actually is

Tuples are immutable sequences; sets are unordered unique collections; dictionaries are key-value maps for labeled data.

When to use it

Tuple: fixed pairs (x,y), function return of multiple values. Set: remove duplicates, membership tests. Dict: records, configs, counting by key.

Where to use it

Dicts for JSON-like data and user profiles; sets for unique visitor IDs; tuples for RGB colors and database rows.

Real use example

A config dict `{"host": "localhost", "port": 8080}` feeds settings into a web app; changing port in one place updates the whole program.

Chapter 7: Strings and Their Handling

Chapter focus: Strings and Their Handling

Strings hold text - names, messages, file paths, or any characters in quotes. Single or double quotes both work. You can combine strings with `+`, repeat with `*`, and slice with `text[start:end]`. Methods like `.upper()`, `.lower()`, `.strip()`, and `.replace()` transform text without writing loops. f-strings (`f"Hello {name}"`) insert variables cleanly. Strings are immutable: methods return new strings rather than changing the original.

Code Reference:

```
name = " python "  
print(name.strip().title())    # Python  
print(f"Hi, {name.strip()}!")
```

What it shows: strip() removes spaces; title() capitalizes; f-strings embed values in text.

Topic guide

What it actually is

A string is an ordered sequence of characters. Python treats it as a sequence, so you can index, slice, and loop over it like a list of letters.

When to use it

Use strings for any text: user input, labels, URLs, CSV fields, error messages, or building output for `print()` and files.

Where to use it

Web forms, log files, data cleaning (trimming spaces), password checks, generating emails, and parsing file names.

Real use example

A program reads ' JOHN DOE ' from a form, uses `.strip().title()` to get 'John Doe', and saves it consistently to a database.

Chapter 8: Reading and Writing Files

Chapter focus: Reading and Writing Files

Use `pathlib.Path` for modern path handling. Check `Path('data.csv').exists()` before reading. `Encoding='utf-8'` avoids character errors on Windows.

Code Reference:

```
with open("notes.txt", "w") as f:  
    f.write("Line 1\nLine 2\n")  
  
with open("notes.txt") as f:  
    print(f.read())
```

What it shows: First block writes two lines; second reads the whole file back.

Topic guide

What it actually is

File handling is reading from or writing to disk. It connects your program's memory to persistent storage.

When to use it

Use files to save logs, load configs, export reports, read CSV datasets, or store user progress.

Where to use it

Data pipelines, backup scripts, game save files, reading homework datasets, generating invoices.

Real use example

A backup script copies every .docx from ~/Documents to ~/Backup using shutil and Path.glob(*.docx').

Chapter 9: Organizing Files

Chapter focus: Organizing Files

Use pathlib.Path for modern path handling. Check Path('data.csv').exists() before reading. Encoding='utf-8' avoids character errors on Windows.

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Chapter 10: Tools for Debugging

Chapter focus: Tools for Debugging

When Python hits a problem it raises an exception and stops unless you handle it. try: runs risky code; except ErrorType: runs if that error happens. Read tracebacks from the bottom up - the last line often says the error type and message. Common errors: ValueError (bad conversion), ZeroDivisionError, FileNotFoundError, KeyError. Handling errors keeps programs friendly instead

of crashing.

Code Reference:

```
try:
    age = int(input("Age: "))
    print(100 / age)
except ValueError:
    print("Please enter a whole number.")
except ZeroDivisionError:
    print("Age cannot be zero.")
```

What it shows: Each except catches a specific error so the user gets a clear message.

Topic guide

What it actually is

An exception is Python's signal that something went wrong at runtime. Handling means catching that signal and recovering gracefully.

When to use it

Use try/except around user input, file access, network calls, and int/float conversions.

Where to use it

Forms, file importers, API clients, calculators, and any code that can receive bad data.

Real use example

A CSV importer tries to open a file; on FileNotFoundError it prints 'File missing' instead of a scary traceback for the end user.

Chapter 11: Tuples

Chapter focus: Tuples

Tuples use parentheses, keep order, and cannot be changed after creation - ideal for coordinates, RGB colors, and database rows returned from queries.

Code Reference:

```
point = (10, 20)
unique = {1, 2, 2, 3}
user = {"name": "Ana", "role": "admin"}
print(point[0], unique, user["name"])
```

What it shows: Tuple index works; set removes duplicate 2; dict lookup uses the key 'name'.

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Tuples are immutable sequences; sets are unordered unique collections; dictionaries are key-value maps for labeled data.

When to use it

Tuple: fixed pairs (x,y), function return of multiple values. Set: remove duplicates, membership tests. Dict: records, configs, counting by key.

Where to use it

Dicts for JSON-like data and user profiles; sets for unique visitor IDs; tuples for RGB colors and database rows.

Real use example

A GPS app stores each waypoint as (latitude, longitude) tuples in a route list that must not be altered accidentally.

Chapter 12: Web Scraping

Chapter focus: Web Scraping

Fetch pages with requests; parse HTML with BeautifulSoup. Respect robots.txt and rate limits.

Code Reference:

```
function greet(name) {  
  return "Hello, " + name;  
}  
console.log(greet("World"));
```

What it shows: A JavaScript function returns a greeting; console.log prints in the browser or Node.

Topic guide

What it actually is

Web programming is creating applications accessed through browsers or HTTP APIs.

When to use it

Websites, online forms, dashboards, REST APIs, interactive tutorials.

Where to use it

Startups, e-commerce, school club sites, internal company tools.

Real use example

A price tracker script reads a product page nightly, parses the price span, and emails if it drops below a target.

Chapter 13: Data and Statistics Management

Chapter focus: Data and Statistics Management

Data science uses programming to collect, clean, analyze, and explain data. The typical flow: ask a question, load data (CSV, database), explore with summaries and charts, find patterns, and share results. Python is popular because libraries like NumPy, Pandas, and Matplotlib handle heavy work. You still need clean data and clear questions - tools alone do not guarantee good insights.

Code Reference:

```
import pandas as pd
df = pd.DataFrame({"sales": [120, 150, 90]})
print(df["sales"].mean())
print(df.describe())
```

What it shows: Pandas loads a table; mean() averages sales; describe() shows count, min, max, and more.

Topic guide

What it actually is

Data science is turning raw data into useful answers using statistics, visualization, and sometimes machine learning.

When to use it

When you have data and need trends, comparisons, forecasts, or evidence for decisions.

Where to use it

Marketing analytics, healthcare research, sports stats, finance, and school science projects.

Real use example

A store loads monthly sales CSV, computes average per product, plots a bar chart, and finds which item dropped 20% - guiding the next order.

Chapter 14: Excel, Charts, Spreadsheets, and Images

Chapter focus: Excel, Charts, Spreadsheets, and Images

openpyxl reads/writes .xlsx files; iterate rows with ws.iter_rows() for spreadsheet automation.

Code Reference:

```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar"]
sales = [120, 150, 90]
plt.bar(months, sales)
plt.title("Monthly Sales")
plt.show()
```

What it shows: bar() draws columns; title labels the chart; show() displays it.

Topic guide

What it actually is

Data visualization is graphical display of data to reveal trends, outliers, and comparisons.

When to use it

Presenting results, exploring data, spotting spikes/drops, reports and slides.

Where to use it

Business dashboards, science papers, journalism, ML training curve monitoring.

Real use example

Monthly reports merge four department Excel files into one summary workbook automatically.